DATTA SABLE / PRACTICAL ANALYTICS SERIES

POWER BI MASTERY

From Raw Data to Executive Dashboards

A hands-on guide to Power Query, data modeling, DAX, visual storytelling, governance, and deployment

Includes a connected practice dataset • 3 guided builds • exercises • interview questions

First Edition | October 2026 | dattasable.com

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All datasets and sample business metrics in this package are synthetic. They are not customer data and should not be represented as real business results.

How to use this book

- Read Chapters 1–3 before building visuals; strong models matter more than decorative charts.
- Keep Power BI Desktop open while completing the labs. Import the four CSV files supplied in the ZIP.
- Type DAX measures rather than only copying them. Verify totals and filter context after each step.
- Complete each chapter checkpoint before advancing. Use the final project rubric to assess readiness.

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The practice case: Northstar Retail

You are a BI analyst supporting Northstar Retail, a fictional multichannel retailer. Leadership wants a trusted view of sales, margin, orders, customer segments, and product performance. The accompanying CSVs form a small but realistic star-schema exercise.

FactSales.csv — Transaction lines: OrderID, OrderDate, CustomerID, ProductID, Quantity, UnitPrice, UnitCost, DiscountPct, Channel, Region.

DimCustomer.csv — CustomerID, CustomerName, Segment, City, State, JoinDate.

DimProduct.csv — ProductID, ProductName, Category, Subcategory, Brand.

DimDate.csv — Date, Year, MonthNo, MonthName, YearMonth, Quarter.

DATA POLICY Values are synthetic. Currency is presented as USD solely for consistent demonstration. You can change formatting to your local currency.

1 The analytics workflow and Power BI ecosystem

LEARNING OUTCOME • Apply the techniques in this chapter to the Northstar Retail practice case.

What Power BI actually does

Power BI converts operational records into an analytical model and then into interactive reports. A dependable solution has a source layer, transformation layer, semantic model, measures, visual layer, and distribution controls. Power BI Desktop is used for authoring; the Power BI service supports publication, collaboration, and scheduled refresh subject to licensing and permissions.

Questions before charts

Start with decisions: Which regions need intervention? Are discounts increasing sales at the expense of margin? Which categories contribute disproportionate profit? Define grain, audience, metric definitions, refresh frequency, and data ownership before drawing charts.

A reliable KPI contract

For every KPI record its name, business definition, formula, exclusions, aggregation, time grain, currency, and owner. Example: Net Sales = sum of line gross sales after line discounts, excluding any separately recorded refunds. Our sample has no refund table, so it is not a net-of-returns measure.

Lab: inspect the CSV package

Open each CSV. Confirm unique ProductID and CustomerID in dimension tables, and multiple rows per customer/product in FactSales. Identify that each FactSales row represents one order line. Write down three questions management could answer.

Checkpoint

Explain the difference between a report, semantic model, and dashboard in the Power BI service. Describe why a KPI definition must be agreed before publishing.

Practice reflection

Write down one business decision this technique enables, one failure mode to watch for, and one validation check you can repeat on a future project.

2 Connect, profile, and clean data in Power Query

LEARNING OUTCOME • Apply the techniques in this chapter to the Northstar Retail practice case.

Load the four files

In Power BI Desktop choose Get data > Text/CSV for each file, then Transform Data. Set FactSales[OrderDate], DimCustomer[JoinDate], and DimDate[Date] to Date; Quantity to Whole Number; UnitPrice, UnitCost, and DiscountPct to Decimal Number. Check locale if decimal parsing differs.

Profile before transformation

Enable Column quality, Column distribution, and Column profile from View. Inspect blanks, errors, distinct values, and outliers. Confirm primary keys in dimension tables are unique. Do not remove legitimate repeated customer IDs from the fact table.

Design repeatable cleaning

Prefer named, ordered transformation steps. Rename cryptic columns, enforce types, trim text where appropriate, and isolate rejected records for review. Avoid silently replacing invalid financial values with zero: doing so changes totals and hides data-quality problems.

Power Query M example

```
let
    Source = Csv.Document(File.Contents("C:\Data\FactSales.csv"), [Delimiter=",", Encoding=65001,
QuoteStyle=QuoteStyle.Csv]),
    Headers = Table.PromoteHeaders(Source, [PromoteAllScalars=true]),
    Typed = Table.TransformColumnTypes(Headers, {{"OrderDate", type date}, {"Quantity",
Int64.Type}, {"UnitPrice", type number}, {"UnitCost", type number}, {"DiscountPct", type
number}})
in
    Typed
```

Checkpoint

Explain why a column with 2% errors deserves investigation before Replace Errors. Identify three transformations that should be documented for auditability.

Practice reflection

Write down one business decision this technique enables, one failure mode to watch for, and one validation check you can repeat on a future project.

3 Model data using a star schema

LEARNING OUTCOME • Apply the techniques in this chapter to the Northstar Retail practice case.

Separate facts from dimensions

FactSales contains measurable events at order-line grain. DimCustomer, DimProduct, and DimDate provide descriptive filters. Create one-to-many relationships from each dimension key to its corresponding FactSales key; use single-direction filtering from dimensions to fact as the default.

Create the relationships

DimCustomer[CustomerID] (1) → FactSales[CustomerID] (*); DimProduct[ProductID] (1) → FactSales[ProductID] (*); DimDate[Date] (1) → FactSales[OrderDate] (*). Ensure data types match, keys are unique on the one side, and no unexpected blank relationship members appear.

Why not one giant flat table?

A star schema improves metric consistency, reduces duplicate descriptions, simplifies slicers, and helps reasoning about filter propagation. A single flat table can be acceptable for tiny prototypes, but grows fragile with multiple facts and reusable dimensions.

Date table checklist

DimDate has one row per calendar date covering the sample range, no gaps or duplicates, and a Date column of type Date. Mark it as the date table when appropriate. Sort MonthName by MonthNo; use YearMonth for chronological trends across years.

Checkpoint

Sketch the four-table model. Explain the meaning of 1:* cardinality and why bi-directional relationships should not be added indiscriminately.

Practice reflection

Write down one business decision this technique enables, one failure mode to watch for, and one validation check you can repeat on a future project.

4 DAX foundations: context, measures, and CALCULATE

LEARNING OUTCOME • Apply the techniques in this chapter to the Northstar Retail practice case.

Measures versus calculated columns

Measures evaluate at query time in the filter context of the visual; calculated columns store row-level results during refresh. Prefer measures for aggregated business KPIs. Use calculated columns only where row-level persisted logic is justified.

Build the core measures

```
Gross Sales = SUMX(FactSales, FactSales[Quantity] * FactSales[UnitPrice])
Discount Amount = SUMX(FactSales, FactSales[Quantity] * FactSales[UnitPrice] *
FactSales[DiscountPct])
Net Sales = [Gross Sales] - [Discount Amount]
Total Cost = SUMX(FactSales, FactSales[Quantity] * FactSales[UnitCost])
Gross Profit = [Net Sales] - [Total Cost]
Gross Margin % = DIVIDE([Gross Profit], [Net Sales])
Orders = DISTINCTCOUNT(FactSales[OrderID])
Units Sold = SUM(FactSales[Quantity])
Average Order Value = DIVIDE([Net Sales], [Orders])
```

Understand filter context

A card with [Net Sales] shows all currently selected transactions. A bar chart by DimProduct[Category] evaluates the same measure once per category. A slicer on Region narrows both. The measure remains unchanged; the filter context changes.

CALCULATE example

```
Online Net Sales = CALCULATE([Net Sales], FactSales[Channel] = "Online")
Online Sales Share = DIVIDE([Online Net Sales], [Net Sales])
```

Common mistakes

Do not sum an already aggregated measure inside SUM. Do not divide without DIVIDE when denominators may be zero. Do not treat SUM of line-level UnitPrice as revenue. Distinct order count differs from row count because an order may have multiple lines.

Checkpoint

Explain why [Net Sales] changes under a Category slicer. Create a measure for Store Net Sales and validate that Online + Store equals Net Sales in the sample.

Practice reflection

Write down one business decision this technique enables, one failure mode to watch for, and one validation check you can repeat on a future project.

5 Time intelligence and business KPIs

LEARNING OUTCOME • Apply the techniques in this chapter to the Northstar Retail practice case.

Prepare a complete calendar

Time intelligence relies on a reliable date dimension and an active relationship to the relevant fact date. Validate the full date range. For rolling comparisons, consider whether missing dates or partial current periods could mislead.

Create date-aware measures

```
Sales YTD = TOTALYTD([Net Sales], DimDate[Date])
Sales Prior Year = CALCULATE([Net Sales], SAMEPERIODLASTYEAR(DimDate[Date]))
YoY Growth % = DIVIDE([Net Sales] - [Sales Prior Year], [Sales Prior Year])
Sales Previous Month = CALCULATE([Net Sales], DATEADD(DimDate[Date], -1, MONTH))
MoM Growth % = DIVIDE([Net Sales] - [Sales Previous Month], [Sales Previous Month])
```

Interpret blank results

The supplied practice data spans two calendar years. Prior-year measures may be blank for periods with no matching history. This is correct and should not automatically be replaced by zero. Use descriptive labels for incomplete periods.

Build a KPI dictionary

Revenue: [Net Sales]. Profit: [Gross Profit]. Margin: [Gross Margin %]. Orders: [Orders]. AOV: [Average Order Value]. Each metric should have one authoritative measure and consistent currency/percentage formatting.

Checkpoint

Place YearMonth on a line chart and compare [Net Sales] to [Sales Prior Year]. Explain what happens when the first month has no comparable prior-year data.

Practice reflection

Write down one business decision this technique enables, one failure mode to watch for, and one validation check you can repeat on a future project.

6 Designing accessible executive dashboards

LEARNING OUTCOME • Apply the techniques in this chapter to the Northstar Retail practice case.

Structure the reading order

Start with a clear title, reporting period, and data freshness label. Follow with 4–6 KPI cards, one main trend, two diagnostic breakdowns, and a concise insight panel. Place detailed tables on drill-through pages rather than squeezing everything onto the overview.

Choose visuals by question

Trend over time → line chart. Compare categories → horizontal bar chart. Composition → stacked bar when the total matters. Distribution → histogram. Geography → map only when geography adds explanatory value. Avoid 3D effects and unnecessary pie charts.

Use a disciplined design system

Use a light neutral canvas, dark navy headings, teal highlights, and one warm warning color. Keep a spacing grid (8/16/24 px), align visual edges, and use no more than two typeface families. Reserve color for meaning, not decoration.

Accessibility is part of quality

Provide sufficient contrast, descriptive titles, alt text where supported, logical tab order, and labels that do not rely only on color. Use readable number formatting: $1.2M, 18.4%, and 12.6K, with clear definitions.

Checkpoint

Create a 16:9 wireframe on paper with six KPI slots, a trend area, category bars, and an insight area. Write a question each visual answers.

Practice reflection

Write down one business decision this technique enables, one failure mode to watch for, and one validation check you can repeat on a future project.

7 Building the Sales Performance report

LEARNING OUTCOME • Apply the techniques in this chapter to the Northstar Retail practice case.

Page 1: Executive Overview

Create cards for Net Sales, Gross Profit, Gross Margin %, Orders, and Average Order Value. Add a YearMonth line chart of Net Sales, Category bar chart of Net Sales, and Region bar chart of Gross Profit. Add slicers for Year, Region, and Channel.

Page 2: Sales Performance

Use a clustered column chart for monthly Net Sales and Gross Profit; a matrix for Region × Category; and a channel comparison bar. Add a tooltip with Units Sold, Orders, and Gross Margin %. Keep units and currency consistent.

Make interactions intentional

Use Edit interactions to control whether slicers filter or highlight each visual. Test clicking a category bar and verify every dependent KPI. Use a Reset Filters bookmark only after ensuring bookmark data settings behave as intended.

Insight writing formula

Observation → comparison → implication → next action. Example: “The West region has the highest revenue but below-average margin; review category mix and discount patterns before expanding promotions.” Validate the observation against your own imported data rather than copying it as a fact.

Checkpoint

Deliver two pages, three slicers, five tested measures, and a one-paragraph executive insight based on your results.

Practice reflection

Write down one business decision this technique enables, one failure mode to watch for, and one validation check you can repeat on a future project.

8 Customer, product, and regional analysis

LEARNING OUTCOME • Apply the techniques in this chapter to the Northstar Retail practice case.

Customer view

Build a segment bar chart, top customers table, and AOV by segment. Consider the distinction between customer count in the dimension and purchasing customers in the fact table. Define Active Customers = DISTINCTCOUNT(FactSales[CustomerID]).

Product view

Rank products by Net Sales and Gross Profit; show margin % alongside absolute profit. A high-revenue product may contribute little profit after discounts. Use a Top N filter carefully and verify ties and grand totals.

Region view

Compare Net Sales, Gross Profit, Orders, and Margin % by Region. Normalize where necessary: large regions naturally have more orders. A high-level regional chart should link to a drill-through page for products and channels.

Additional measures

```
Active Customers = DISTINCTCOUNT(FactSales[CustomerID])
Revenue per Customer = DIVIDE([Net Sales], [Active Customers])
Profit per Order = DIVIDE([Gross Profit], [Orders])
Discount Rate % = DIVIDE([Discount Amount], [Gross Sales])
```

Checkpoint

Identify one product with strong sales but weak margin. State whether its margin issue is driven by price, cost, or discount; use the data to support your reasoning.

Practice reflection

Write down one business decision this technique enables, one failure mode to watch for, and one validation check you can repeat on a future project.

9 Performance, security, publishing, and governance

LEARNING OUTCOME • Apply the techniques in this chapter to the Northstar Retail practice case.

Performance first principles

Remove unused columns, favor a star schema, avoid unnecessarily high-cardinality text columns, and minimize expensive visuals. Use Performance Analyzer to identify slow visuals and measure changes rather than guessing. Import mode is suitable for this small CSV practice case.

Row-level security basics

RLS restricts rows visible to viewers when correctly configured in the semantic model and service. It is not a substitute for controlling workspace roles, access to source files, or sensitive exports. Test roles with View as and verify behavior in the service.

Publish responsibly

Publish to a suitable workspace, configure credentials and refresh where needed, and check who can access the report and underlying semantic model. Confirm licensing, sharing permissions, and organizational data policy before external distribution.

Quality gates

Validate model cardinality, measure totals, date filtering, slicer interactions, accessibility, refresh results, export behavior, and viewer permissions. Record dataset version, metric definitions, and known limitations in a release note.

Checkpoint

List five things you would test before sending an executive report link. Explain why hiding a page does not secure the data behind it.

Practice reflection

Write down one business decision this technique enables, one failure mode to watch for, and one validation check you can repeat on a future project.

10 Capstone project and interview preparation

LEARNING OUTCOME • Apply the techniques in this chapter to the Northstar Retail practice case.

Your assignment

Build a three-page retail analytics report: Executive Overview, Product & Margin, and Customer & Region. Use the supplied four-table model, reusable measures, a consistent design system, and a written executive summary.

Acceptance criteria

A passing submission has no broken relationships; at least 10 correct measures; a working date slicer; no misleading percentage aggregation; accessible titles and contrast; and a one-page explanation of findings, caveats, and recommended actions.

Suggested assessment rubric

Data quality and model: 25 points. DAX accuracy: 25. Visual design and interactions: 20. Insight quality: 20. Governance and documentation: 10. Total: 100. A score of 85+ indicates a strong portfolio-ready demonstration, not a professional certification.

Interview practice

Q: Why star schema? A: Clear dimensional filtering and consistent measures. Q: SUM vs SUMX? A: SUM aggregates one numeric column; SUMX evaluates an expression per row. Q: Why use DIVIDE? A: Safe handling of zero/blank denominators. Q: Why does a total differ from sum of visible row percentages? A: Measures recalculate in total filter context. Q: How do you diagnose slow reports? A: Performance Analyzer, model inspection, and targeted DAX optimization.

Final delivery checklist

Save the PBIX locally, export a PDF for review, include a short README with data sources and definitions, and remove any credentials or private source paths before sharing.

Practice reflection

Write down one business decision this technique enables, one failure mode to watch for, and one validation check you can repeat on a future project.

Appendix A DAX quick reference

SUM — Sum a numeric column.

SUMX — Evaluate an expression row by row, then sum.

DISTINCTCOUNT — Count distinct values in current context.

DIVIDE — Divide with safe zero/blank behavior.

CALCULATE — Evaluate an expression in modified filter context.

FILTER — Return a table filtered by an expression.

ALL — Remove filters from specified table or columns; use carefully.

REMOVEFILTERS — Explicitly clear filters in CALCULATE.

TOTALYTD — Year-to-date evaluation with a date column.

SAMEPERIODLASTYEAR — Shift date context to prior year.

Appendix B Essential glossary

Grain — What a single fact-table row represents.

Dimension — Descriptive table used to filter and group facts.

Fact — Table of events and numeric measurements.

Filter context — Set of active filters under which a measure evaluates.

Row context — Current row used during row-by-row evaluation.

Measure — DAX expression evaluated dynamically in context.

Semantic model — Relationships, tables, measures, and metadata supporting reporting.

RLS — Row-level security rules limiting visible data.

Appendix C Pre-sale and publication quality checklist

- Proofread all pages and check that page numbers and headings render correctly.
- Test every DAX example in Power BI Desktop with the supplied CSVs.
- Ensure every external reference is linked and not copied without permission.
- Add your store terms, support contact, refund policy, and applicable tax information before listing.
- Test download ZIP on Windows and mobile; verify CSV UTF-8 encoding and filenames.
- Review Microsoft licensing and feature documentation for changes before each new edition.

Official further reading

Microsoft Learn: https://learn.microsoft.com/power-bi/
DAX documentation: https://learn.microsoft.com/dax/
Power Query documentation: https://learn.microsoft.com/power-query/
Power BI guidance: https://learn.microsoft.com/power-bi/guidance/

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